JOINT ADMISSION TEST FOR MASTERS • IIT JAM MATHEMATICS (MA)

IIT JAM 2012 Official Paper

Category: Classic Era (2005–2014) • Official Mock Test Series

TOTAL QUESTIONS	TOTAL MARKS	TEST DURATION	PAPER PATTERN
15	90 M	60 Min	MCQ

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	15	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)

IMPORTANT INSTRUCTIONS FOR CANDIDATES

1. **Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
2. **Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
3. **Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
4. **Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
5. **Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
6. **Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at **vaibhavgeometer.github.io**.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2012 • Q1 • ID: JAM_2012_Q1) MCQ +6 M Neg: −2.0

Q.1 Let $\{x_n\}$ be the sequence $+\sqrt{1}, -\sqrt{1}, +\sqrt{2}, -\sqrt{2}, +\sqrt{3}, -\sqrt{3}, +\sqrt{4}, -\sqrt{4}, \ldots$.
If

$$y_n = \frac{x_1 + x_2 + \cdots + x_n}{n} \text{ for all } n \in \mathbb{N},$$

then the sequence $\{y_n\}$ is

(A) monotonic
(B) NOT bounded
(C) bounded but NOT convergent
(D) convergent

Question 2 (JAM 2012 • Q2 • ID: JAM_2012_Q2) MCQ +6 M Neg: −2.0

Q.2 The number of distinct real roots of the equation $x^9 + x^7 + x^5 + x^3 + x + 1 = 0$ is

(A) 1 (B) 3 (C) 5 (D) 9

Question 3 (JAM 2012 • Q3 • ID: JAM_2012_Q3) MCQ +6 M Neg: −2.0

Q.3 If $f : \mathbb{R}^2 \to \mathbb{R}$ is defined by

$$f(x,y) = \begin{cases} \dfrac{x^3}{x^2 + y^4} & \text{if } (x,y) \neq (0,0), \\ 0 & \text{if } (x,y) = (0,0), \end{cases}$$

then

(A) $f_x(0,0) = 0$ and $f_y(0,0) = 0$
(B) $f_x(0,0) = 1$ and $f_y(0,0) = 0$
(C) $f_x(0,0) = 0$ and $f_y(0,0) = 1$
(D) $f_x(0,0) = 1$ and $f_y(0,0) = 1$

Question 4 (JAM 2012 • Q4 • ID: JAM_2012_Q4) MCQ +6 M Neg: −2.0

Q.4 The value of $\int_{z=0}^{1}\int_{y=0}^{z}\int_{x=0}^{y} x\,y^2 z^3\,dx\,dy\,dz$ is

(A) $\frac{1}{90}$ (B) $\frac{1}{50}$ (C) $\frac{1}{45}$ (D) $\frac{1}{10}$

MA- 1 / 36

Question 5 (JAM 2012 • Q5 • ID: JAM_2012_Q5) MCQ +6 M Neg: −2.0

Q.5 The differential equation $\left(1+x^2y^3+\alpha x^2y^2\right)dx+\left(2+x^3y^2+x^3y\right)dy=0$ is exact if α equals

(A) $\frac{1}{2}$ (B) $\frac{3}{2}$ (C) 2 (D) 3

Question 6 (JAM 2012 • Q6 • ID: JAM_2012_Q6) MCQ +6 M Neg: −2.0

Q.6 An integrating factor for the differential equation $\left(2xy+3x^2y+6y^3\right)dx+\left(x^2+6y^2\right)dy=0$ is

(A) x^3 (B) y^3 (C) e^{3x} (D) e^{3y}

Question 7 (JAM 2012 • Q7 • ID: JAM_2012_Q7) MCQ +6 M Neg: −2.0

Q.7 For $c>0$, if $a\hat{i}+b\hat{j}+c\hat{k}$ is the unit normal vector at $\left(1,1,\sqrt{2}\right)$ to the cone $z=\sqrt{x^2+y^2}$, then

(A) $a^2+b^2-c^2=0$ (B) $a^2-b^2+c^2=0$

(C) $-a^2+b^2+c^2=0$ (D) $a^2+b^2+c^2=0$

Question 8 (JAM 2012 • Q8 • ID: JAM_2012_Q8) MCQ +6 M Neg: −2.0

Q.8 Consider the quotient group $\mathbb{Q}/\mathbb{Z}$ of the additive group of rational numbers. The order of the element $\frac{2}{3}+\mathbb{Z}$ in $\mathbb{Q}/\mathbb{Z}$ is

(A) 2 (B) 3 (C) 5 (D) 6

Question 9 (JAM 2012 • Q9 • ID: JAM_2012_Q9) MCQ +6 M Neg: −2.0

Q.9 Which one of the following is TRUE ?

(A) The characteristic of the ring $6\mathbb{Z}$ is 6
(B) The ring $6\mathbb{Z}$ has a zero divisor
(C) The characteristic of the ring $(\mathbb{Z}/6\mathbb{Z})\times 6\mathbb{Z}$ is zero
(D) The ring $6\mathbb{Z}\times 6\mathbb{Z}$ is an integral domain

Question 10 (JAM 2012 • Q10 • ID: JAM_2012_Q10) MCQ +6 M Neg: −2.0

Q.10 Let W be a vector space over $\mathbb{R}$ and let $T:\mathbb{R}^6\to W$ be a linear transformation such that $S=\{Te_2, Te_4, Te_6\}$ spans W. Which one of the following must be TRUE ?

(A) S is a basis of W
(B) $T(\mathbb{R}^6)\neq W$
(C) $\{Te_1, Te_3, Te_5\}$ spans W
(D) $\ker(T)$ contains more than one element

MA- 2 / 36

Question 11 (JAM 2012 • Q11 • ID: JAM_2012_Q11) MCQ +6 M Neg: −2.0

Q.11 Consider the following subspace of $\mathbb{R}^3$:

$$W = \{(x, y, z) \in \mathbb{R}^3 \mid 2x + 2y + z = 0,\ 3x + 3y - 2z = 0,\ x + y - 3z = 0\}.$$

The dimension of W is

(A) 0 (B) 1 (C) 2 (D) 3

Question 12 (JAM 2012 • Q12 • ID: JAM_2012_Q12) MCQ +6 M Neg: −2.0

Q.12 Let P be a 4×4 matrix whose determinant is 10. The determinant of the matrix $-3P$ is

(A) -810 (B) -30 (C) 30 (D) 810

Question 13 (JAM 2012 • Q13 • ID: JAM_2012_Q13) MCQ +6 M Neg: −2.0

Q.13 If the power series $\sum_{n=0}^{\infty} a_n x^n$ converges for $x = 3$, then the series $\sum_{n=0}^{\infty} a_n x^n$

(A) converges absolutely for $x = -2$
(B) converges but not absolutely for $x = -1$
(C) converges but not absolutely for $x = 1$
(D) diverges for $x = -2$

Question 14 (JAM 2012 • Q14 • ID: JAM_2012_Q14) MCQ +6 M Neg: −2.0

Q.14 If $Y = \left\{ \frac{x}{1+|x|} \,\middle|\, x \in \mathbb{R} \right\}$, then the set of all limit points of Y is

(A) $(-1,1)$ (B) $(-1,1]$ (C) $[0,1]$ (D) $[-1,1]$

Question 15 (JAM 2012 • Q15 • ID: JAM_2012_Q15) **MCQ** **+6 M** **Neg: −2.0**

Q.15 If C is a smooth curve in $\mathbb{R}^3$ from $(0,0,0)$ to $(2,1,-1)$, then the value of

$$\int_C (2xy+z)\,dx + (z+x^2)\,dy + (x+y)\,dz$$

is

(A) -1 (B) 0 (C) 1 (D) 2

MA- 3 / 36

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

IIT JAM 2012 Official Paper • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	MCQ	+6	**D**
2	MCQ	+6	**A**
3	MCQ	+6	**B**
4	MCQ	+6	**A**
5	MCQ	+6	**B**
6	MCQ	+6	**C**
7	MCQ	+6	**A**
8	MCQ	+6	**B**

Q. No.	Type	Marks	Official Key
9	MCQ	+6	**C**
10	MCQ	+6	**D**
11	MCQ	+6	**B**
12	MCQ	+6	**D**
13	MCQ	+6	**A**
14	MCQ	+6	**D**
15	MCQ	+6	**C**

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt (-1/3 for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., [a, b]) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.